

# NEGATIVE FEEDBACK LOOPS OF CHEMICAL KINETICS IN LIQUID-LIQUID PHASE SEPARATION

BACHELOR THESIS

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## **Abstract**

Liquid-liquid phase separation (LLPS) plays a crucial role in organizing biochemical processes in living cells, yet its impact on the dynamics of chemical feedback loops remains poorly understood. In this project, we investigate how negative feedback loops behave in phase-separated environments using both analytical models and numerical simulations based on the Flory-Huggins theory. Three models of chemical kinetics are considered: one assuming a well-mixed system, and two incorporating spatial effects either through first-passage kinetics or surface reaction dynamics. Simulations are performed using finite difference methods to capture the spatial-temporal evolution of concentrations under LLPS. Our results show that while none of the models produce sustained oscillations for two species, phase separation significantly alters the system's fixed-point behavior. Notably, analytical predictions diverge from simulation results, revealing limitations of local mass-action assumptions. These findings suggest that spatial structure and reaction locality must be carefully accounted for to understand feedback regulation in biologically realistic, phase-separated systems. Future work should explore multi-species systems, where sustained oscillations may emerge more readily.

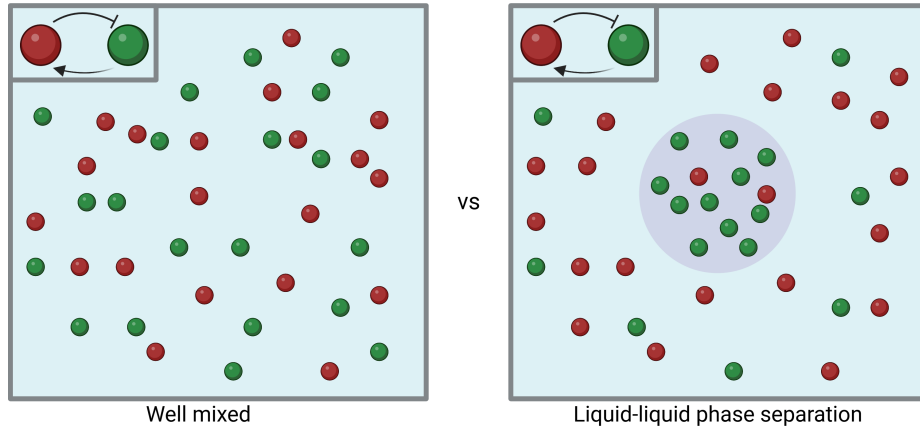
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# 1 Introduction



**Figure 1:** The top left shows the structure of the negative feedback loop between the two species where the regular arrow symbolizes a positive link, and the flat arrow indicates a negative link. The left hand panel shows a well mixed case, and the right hand panel shows a LLPS (liquid-liquid phase separation) case

Chemical kinetics inside the living cell has historically been modeled by assuming a well-mixed environment. Given such assumptions, first-order reactions occur at rates best described by applying the mass action law, which simply states that a given reaction happens at a rate proportional to the concentrations of reactants:  $r_{aA+bB+\dots \rightarrow Products} = k \cdot [A]^a \cdot [B]^b \cdot \dots$ . In the case of a diffusion-limited reaction with two reactants,  $k$  can be calculated using the Smoluchowski equation. [1]

However, recent discoveries suggest that the inside of a cell, in addition to containing the classical organelles, is subject to LLPS, resulting in regions of different compositions, sometimes referred to as membrane-less organelles[2]. Phase separation occurs when some chemicals are more inclined to group together or repel each other, resulting in stable regions of qualitatively different compositions such as water separating from oil.[3]

We aim to describe how phase separation affects chemical kinetics, with emphasis on negative feedback loops. Feedback loops are systems in which some variable, in our case concentrations, affect their own future value, either directly or indirectly. In the case of negative feedback loops, the value of the variable negatively affects its future value. One such negative feedback loop in which phase separation has been proposed to influence dynamics is the p53 pathway where the DNA-repairing protein p53 forms a negative feedback loop with the mdm2 regulator[4], which initially prompted a closer investigation of how chemical dynamics behave in a phase separated environment.

To do this spatial interactions are generated by using finite difference methods to simulate a system governed by the Flory-Huggins chemical potential. The chemical

reactions are modeled by locally letting the mass-action law govern reactions. We compare the overall mean chemical concentrations generated by spatial simulations to 2 proposed analytical models. Both models assume chemicals separating from the surroundings into one droplet. One model assumes a constant reaction rate on the surface of the droplet, and one assumes that reactions happen according to the first passage rate of a particle finding the droplet.

We discover a discrepancy between analytical models and the simulation, and discuss what the reason for this might be.

## 1.1 Negative Feedback Loops

The essence of this project is to study negative feedback loops. Feedback loops are systems in which the output of the system is used as input in the system for future operations. Suppose that we have a network consisting of  $N$  nodes, where the value of node  $i$  influences the rate of change of node  $i + 1$  and the value of node  $N$  influences the rate of change of node 1. In this way each node indirectly affects itself. The interaction links can be positive, that is, node  $i$  increases production or reduces the degradation of node  $i + 1$ , or they can be negative where node  $i$  inhibits production or degrades node  $i + 1$ .

If the total amount of negative links in the loop is uneven the loop is said to be negative, as a node will have a negative effect on its own future value. The loop is positive for an even number of negative links. The generic formula for inhibition of production is  $\frac{dx_{i+1}}{dt} = A \cdot \frac{f(x_i)}{K+f(x_i)}$  and degradation can be written as  $\frac{dx_{i+1}}{dt} = -A \cdot x_{i+1}f(x_i)$ . Here  $f$  is an arbitrary growing function,  $A$  is a baseline degradation or growth rate, which itself might be a function of the state of the system and  $K$  is some constant.

## 1.2 Liquid phase separation & Flory-Huggins

Negative feedback loops in well-mixed environments are well described and thus we wish to study the dynamics of such loops in systems with phase separation. Phase separation occurs when chemicals spatially organize themselves into regions with different properties; this can be the physical structure such as liquid vs. solid, but it can also be regions with certain concentrations or compositions. An everyday example of this could be that oil separates from water.

A mathematical model for phase separation is needed. The basic idea of the model is that entropy is driving the system towards the well-mixed case, but when some molecules attract or repel each other these neighbor interactions are in some cases sufficient to overcome the entropy.

### 1.2.1 Neighbor interactions

Suppose all the molecules are organized in a grid. If there are 3 molecule species A, B and C, with  $M = N_A + N_B + N_C$  being the total number of molecules, then all molecules will have the same volume  $\nu$ .

The energy associated with neighbor interactions can be denoted  $e_{ij}$ , where  $i, j \in \{A, B, C\}$ . E.g. if molecule A and molecule B are next to each other, the corresponding energy will be  $e_{AB}$ . In the continuum limit, and by applying the Sterling and mean-field approximations we arrive at the following expression for the Helmholtz free energy (A full derivation is provided in appendix A).

$$F(\phi_A, \phi_B) = F_0 + \int dr^3 \frac{1}{\nu} \left( k_B T \cdot \left[ \phi_A \ln \phi_A + \phi_B \ln \phi_B + (1 - \phi_A - \phi_B) \ln(1 - \phi_A - \phi_B) \right. \right. \\ \left. \left. + \chi_{AB} \phi_A \phi_B + (\chi_{AC} \phi_A + \chi_{BC} \phi_B)(1 - \phi_A - \phi_B) \right. \right. \\ \left. \left. - \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \nabla \phi_A \nabla \phi_B - \chi_{BC} |\nabla \phi_B|^2 - \chi_{AC} |\nabla \phi_A|^2 \right) \right] \right) \quad (1)$$

This is the so-called Flory-Huggins potential.[5] The Flory-Huggins parameter  $\chi_{ij} = \frac{z(2e_{ij} - e_{ii} - e_{jj})}{k_B T}$  is a quantization of how much two chemicals repel each other. Due to the second law of thermodynamics a system will try to minimize its free energy. It does so by varying the functional, inside the integral, so expressions for the chemical potentials of exchange are obtained by taking the functional derivative,  $\frac{\delta F}{\delta \phi_i} = \frac{\partial f}{\partial \phi_i} - \nabla \cdot \left( \frac{\partial f}{\partial \nabla \phi_i} \right)$ :

$$\mu_A = \nu \frac{\delta F}{\delta \phi_A} = \ln \frac{\phi_A}{1 - \phi_A - \phi_B} + (\chi_{AB} - \chi_{BC}) \phi_B + \chi_{AC} (1 - 2\phi_A - \phi_B) \\ + \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \Delta \phi_B - 2\chi_{AC} \Delta \phi_A \right) \quad (2)$$

$$\mu_B = \nu \frac{\delta F}{\delta \phi_B} = \ln \frac{\phi_B}{1 - \phi_A - \phi_B} + (\chi_{AB} - \chi_{AC}) \phi_A + \chi_{BC} (1 - \phi_A - 2\phi_B) \\ + \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \Delta \phi_A - 2\chi_{BC} \Delta \phi_B \right) \quad (3)$$

The logarithmic terms in front correspond to the entropy density, the linear terms on the right of the first lines represent the energy of mixing, and finally the terms involving the Laplacian punish spatial inhomogeneities, effectively creating surface tension. These are two of the equations that govern the spatial dynamics of the system.

### 1.3 Temporal evolution

The chemical potentials do not directly say anything about the temporal evolution of the concentrations, however they are necessary to calculate the current,  $J_i$ , of a species  $i$ :

$$J_i = \vec{\nabla} c_i + \Lambda \nabla \mu_i \quad (4)$$

Here the first term is the advection and the second term arises from the thermodynamic force exerted by the chemical potential.  $\Lambda$  is a function describing the molecular mobility, which is a quantization of how easily particles are moved by the thermodynamic forces. For simplicity we will assume constant mobility. The rate of change of the volume fraction of a species in a point must be the divergence of the flux of that species through that point per unit volume:

$$\partial_t \phi_i = \partial_t \frac{c_i}{\nu} = \nabla \cdot J_i = \nabla \cdot \left( \vec{\nabla} \phi_i + \frac{\Lambda}{\nu} \nabla \mu_i \right) \quad (5)$$

Choosing a frame of reference stationary relatively to the liquid and defining  $\tilde{\Lambda} = \frac{\Lambda}{\nu}$  reduces this to:

$$\partial_t \phi_i = \tilde{\Lambda} \Delta \mu_i \quad (6)$$

The equations (2) , (3) & (6) describe the spatial dynamics system and are simulated by integrating them in time. An implication of the equations is that droplets will grow due to two main effects; Ostwald ripening and coalescence. These two effects are characteristic of the system and are a way of testing if the spatial dynamics behave according to theory. The following chapter provides a more in depth explanation of these two effects.

### 1.4 Ostwald ripening and coalescence

In a system with no chemical kinetics the spatial dynamics, governed by (2) and (3), will cause droplets to form and grow. Larger droplets will dominate and grow at the expense of smaller droplets both due to coalescence and Ostwald ripening. In this section we express the expected rate at which this happens. Ostwald ripening is a phenomenon where larger droplets grow and seemingly consume smaller droplets from afar by causing them to dissolve. This happens due to the larger droplets impact on the local environment around the smaller droplets making their phase separation non-maintainable and causing them to dissolve. For a system large enough that  $V \gg \sum_i V_i$ ,  $V_i$  being the size of each droplet, and with enough droplets to be described by a continuous droplet size distribution, we expect the following scaling of average droplet size:

$$\langle R(t) \rangle \propto \left( \frac{D \ell_\gamma c_{out}^{(0)}}{c_{in}^{(0)}} t \right)^{\frac{1}{3}} \quad (7)$$



Where  $c_{in}$  is the concentration inside the droplet, which is assumed to be constant,  $c_{out}$  is the concentration just outside the droplet and  $\ell_\gamma$  is the capillary length. Here  $D$  is the diffusion constant  $D \simeq 4(\chi - 2)\nu\Lambda k_b T$ . This is the Lifshitz-Slyozov scaling law, which dictates that the average droplet radius scales proportional to the cubic root of the time.[6]

It just so happens that the growth rate caused by coalescence is also proportional to  $\sqrt[3]{t}$ . [7] In our system, where both coalescence and Ostwald ripening is expected to occur, we expect the mean droplet size scale with this factor  $\sqrt[3]{t}$ .

## 2 Method

We have now established the mathematical framework and need a way to simulate this. The temporal derivative, Eq. (6), given the chemical potentials described in eqs. (2) and (3) is simulated by calculating spatial derivatives using finite difference grids. These are then used to calculate the temporal evolution using Euler or Runge-Kutta integration.

The chemical reactions are simulated by applying the mass-action locally, which is done by calculating the chemical reaction rates at each grid-point and integrating in time, again using either Euler or Runge-Kutta.

### 2.1 Chemical reactions

We want to model a simple negative 2-species feedback loop. Here we can choose between 2 models that each have advantages and drawbacks.

The first way to model it is:

$$\begin{aligned}\frac{dc_1}{dt} &= a_1 - b_1 \cdot c_1 \cdot c_2 \\ \frac{dc_2}{dt} &= a_2 \cdot c_1 - b_2 \cdot c_2\end{aligned}\tag{8}$$

with 2 fixed points:

$$(c_1^*, c_2^*) = \left( \pm \sqrt{\frac{a_1 b_2}{b_1 a_2}}, \pm \sqrt{\frac{a_1 a_2}{b_1 b_2}} \right)\tag{9}$$

This model would be the obvious choice for modeling a simple 2-species feed-back loop when species 1 is produced at constant rate and degraded by species 2. Species 2 is on the other hand is produced at a rate proportional to the amount of species 1, and degraded at a rate proportional to itself.

The simplicity of this model makes it more manageable to derive the implications of how one would expect it to behave given phase separation. The issue with the model is that it doesn't enforce incompressibility which is necessary for the Flory-Huggins

potential to be well defined.

To get around this problem we propose to model it the following way:

$$\begin{aligned}\frac{dc_1}{dt} &= a_1 \cdot c_S - b_1 \cdot c_2 \cdot c_1 \\ \frac{dc_2}{dt} &= a_2 \cdot c_1 \cdot c_S - b_2 \cdot c_2 \\ c_S &= \frac{1}{\nu} - c_1 - c_2\end{aligned}\tag{10}$$

with fixed points:

$$(c_1^*, c_2^*) \in \left\{ (1, 0); \left( \sqrt{\frac{a_1 b_2}{b_1 a_2}}, \frac{1 - c_1^*}{\frac{b_1}{a_1} c_1^* + 1} \right) \right\}\tag{11}$$

This model does enforce incompressibility as a solvent particle  $S$  is required for the creation of each kind of chemical. It has slightly different dynamics than (8), but in the limit  $c_1, c_2 \ll c_S$  (10) turns into (8). We have chosen to proceed with the latter model for simulations as it is compatible with the Flory-Huggins potential, but it also makes the solvent play an active part in the chemical reactions which is not the case in the actual systems we are trying to model. We have chosen the first model for analytical derivations as it reflects chemical dynamics in the cell better, and it makes derivations more manageable. Switching between the two shouldn't present too big of an issue as, they in many ways display similar dynamics. One such dynamic is the tendency to drive systems towards a certain fixed point, which without spatial interactions leads to a homogeneous environment.

## 2.2 Chemical reactions driving system towards homogeneity

If no spatial interactions are present the system will be driven towards the fixed-point everywhere. To explore this we determine the linear stability of the fixed points thus determining if they are stable and attract. The Jacobian of eq. (10) reads:

$$J = \begin{pmatrix} -a_1 - b_1 \cdot c_2 & -a_1 - b_1 c_1 \\ a_2 \cdot (c_S - c_1) & -a_2 c_1 - b_2 \end{pmatrix}\tag{12}$$

The trace of the Jacobian is always negative which implies that the fixed point is either stable or a saddle point. As the concentrations are limited to the unit square, and this is the only fixed point in this region we conclude that it must be a stable fixed point[8].

Thus if we let the chemical reactions dominate the system, the system will be driven towards the fixed point resulting in a homogeneous system with concentrations  $c_A^*$  and  $c_B^*$  everywhere. For this reason we will assume that chemical reactions occur on a slower timescale than spatial interactions.

### 2.2.1 Timescales

The rate at which the chemical reactions drive the system towards (11) depends on the reaction rates, and as long as  $\frac{b_1}{a_1}$  and  $\frac{a_2}{b_2}$  are constant, the local chemical dynamics remain the same. Thus by scaling all  $a$ 's and  $b$ 's by some constant  $k$ , we can change the reaction speed without changing the dynamics.

The timescale at which the phase separation reaches equilibrium is determined by the mobility constant  $\Lambda$  and the size of the system, because in a larger system particles have to travel a longer distance to find a certain phase.

The exact ratio of  $\Lambda$  to  $k$  was found by trial and error, although qualitatively we can deduce that  $\Lambda$  should be greater than  $k$  by a factor that scales with the system size.

### 2.3 Simulation & Finite difference methods

To solve the differential equations governing the spatial dynamics of the systems, we use FDM (finite difference method). Finite difference is a collection of numerical methods employing Taylor approximations to numerically estimate the derivatives of a function. In one dimension it works by solving the the following matrix equation for  $\vec{w}$ :

$$\begin{pmatrix} (x_1 - x_i)^0 & \cdots & (x_N - x_i)^0 \\ \vdots & & \vdots \\ (x_1 - x_i)^N & \cdots & (x_N - x_i)^N \end{pmatrix} \times \vec{w} = d! \begin{pmatrix} \delta_{0,d} \\ \vdots \\ \delta_{N-1,d} \end{pmatrix} \quad (13)$$

Here  $d$  is the order of derivative in which we are interested, and  $i$  is the index of the point at which we wish to evaluate the derivative, and  $\delta$  is the Kronecker delta. From this it follows that to approximate the  $d$ 'th derivative of a point we need at least  $d+1$  points.

The derivative is then calculated by:

$$\left. \frac{\partial^d f}{\partial x^d} \right|_{x=x_i} \approx \vec{x} \cdot \vec{w} \quad (14)$$

By using a cubic grid we can calculate derivatives along each axis separately, which is the approach we have chosen.[9] As we do not expect a certain fixed region in space to require a higher resolution than others we have used the approach of a regular cubic grid, as opposed to an irregular grid, which would add complexity to the model without providing further insights of the modeled system.

In our code FDM is implemented such that it generalizes to grids in arbitrarily many dimensions by calculating derivatives along each axis separately. The code uses  $d+1$  points to calculate the  $d$ 'th derivative by default. However, it is implemented in the code that this number may be increased if we wish to increase accuracy. It is implemented in a way such that we require uniform spatial resolution along any axis, two

different axes are allowed to have different spatial resolutions.

### 2.3.1 Timesteps

As the systems we are simulating are dynamic and very volatile at some points in time and more stable in others we have incorporated dynamic timesteps. This allows the program to adjust the  $dt$  to an appropriate value for every timestep. This is done by having the program test the value of the  $dt$  it last used, initially set to 0.001, by performing a timestep of size  $dt$  to the state  $F$  and then comparing this to the result of performing two time steps of size  $dt/2$ . This comparison is done on every single grid point instead of the grid as a whole, so as to set the timescale according to the most volatile local part of the system. We set the upper limit of the difference between the results to  $10^{-5}$ . If this limit is not exceeded the state of the system after the two  $dt/2$  timesteps is accepted and we set  $dt_{n+1} = dt_n/0.8$ . If these limits are exceeded we set  $dt_{n+1} = dt_n \cdot 0.8$  and repeat until the  $dt$  is accepted.

### 2.3.2 Boundary conditions

The program incorporates periodic boundary conditions by letting  $f_{N+i} = f_i$ , where  $N$  is the number of gridpoints along the axis of interest. We have also implemented Neumann boundary conditions by using the fact that  $f^{n+1}(x) \approx \frac{f^{(n)}(x+\Delta x) - f^{(n)}(x-\Delta x)}{2\Delta x}$  so at the boundary, either  $f^{(n)}(x - \Delta x)$  or  $f^{(n)}(x + \Delta x)$  might be held constant depending on the given border.

Dirichlet boundary conditions are enforced by updating the value of the grid at the borders to the value of choice.

The code has been implemented such that it is possible to choose what boundary conditions we wish along each axis through parameters.

## 2.4 Parameters

Our program uses a number of parameters. The implications (in practice) of some of these parameters are of the same nature, namely setting the timescale. This includes the mobility coefficient  $\Lambda$ , the molecular volume  $\nu$ ,  $k_b T$  and the list of constants in the chemical reactions. To simplify this we set  $\Lambda, \nu, k_b T = 1$  and use the list  $[a_2, b_2, b_1, a_1]$  of constants linked to the chemical reactions to set the relative timescale of the interactions. The Flory-Huggins interaction parameters  $[\chi_{AB}, \chi_{AC}, \chi_{BC}]$  are also given as input parameters. In (Weber et al. et al., 2019) [7] they choose to neglect the  $\nabla\phi_A\nabla\phi_B$  cross term in the Flory-Huggins free energy expression - equivalent to 1. We, however, chose not to neglect this term and include it in our program. This made for the following limitations on our parameters:  $\chi_{AC} + \chi_{AC} > 0$ . Finally the program uses initial conditions as a parameter.

The *initial conditions* for our simulations with chemical kinetics is a grid with a single droplet. This grid is obtained by setting up a pseudo-random grid with predetermined average concentrations  $\bar{c}_a$  and  $\bar{c}_b$  and standard deviations  $\sigma_a$  and  $\sigma_b$ . This grid is then regularized to ensure that no volume fraction exceeds 1. We then run a simulation on this grid without chemical reactions for an extended amount of time until the system has stabilized with only 1 droplet remaining. The volume of this droplet grows roughly proportional to  $\frac{c_A}{c_B + c_A}$ . With this we now have a framework for how the simulations will be done. A typical simulation might run with the following parameters.

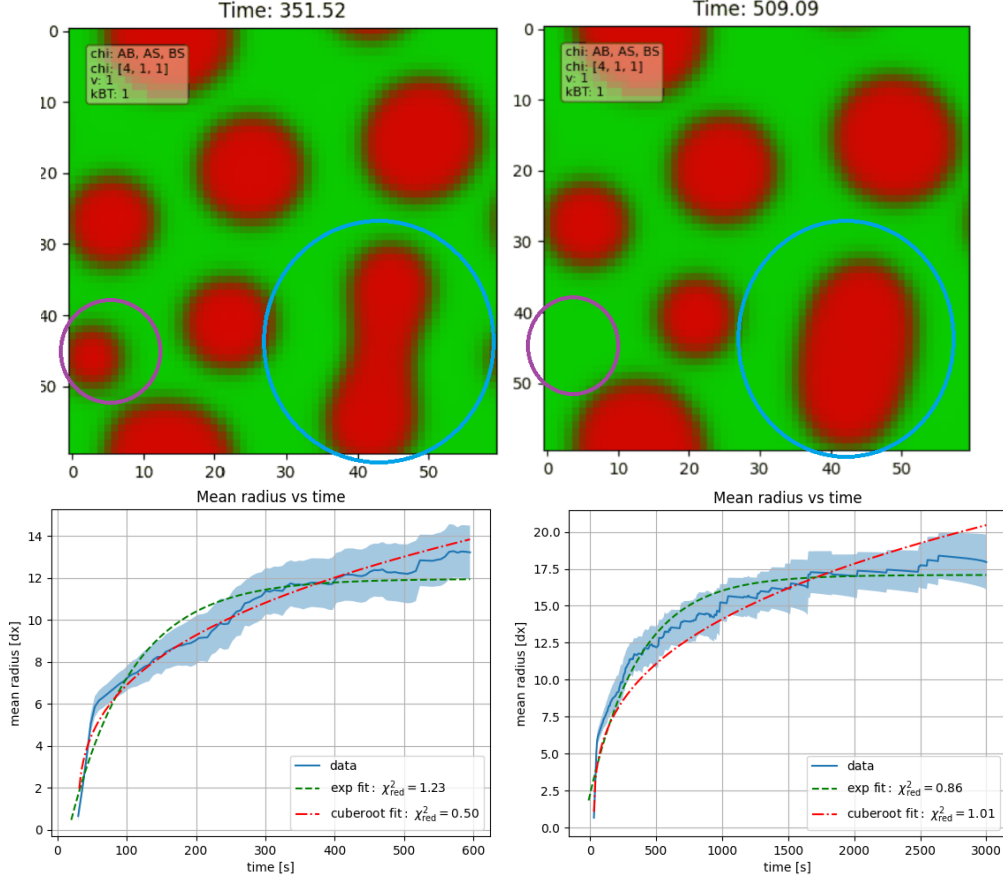
| Gridsize | $\Delta x$ | $\nu$ | $k_b T$ | $[\chi_{AB}, \chi_{AC}, \chi_{BC}]$ | $[a_1, a_2, b_1, b_2]$   |
|----------|------------|-------|---------|-------------------------------------|--------------------------|
| 30x30    | 1          | 1     | 1       | [4, 1, 1]                           | [0.24, 0.08, 0.64, 0.04] |

**Table 1:** Typical list of parameters for a simulation of eq. (10) with a pseudo-random grid as initial conditions.

### 3 Results

#### 3.1 Recovering the Lifshitz-Slyozov scaling law

To test if the simulations are consistent with theory we look for some of the phenomena that should follow from the Flory-Huggins. Namely Coalescence and Ostwald Ripening. In fig. 2 it is seen that the mean droplet radius fits a cube-root function pretty well on a short time-scale, but when approaching the maximum droplet size due to the limited size of the system, an exponential decay-like function describes the data better.

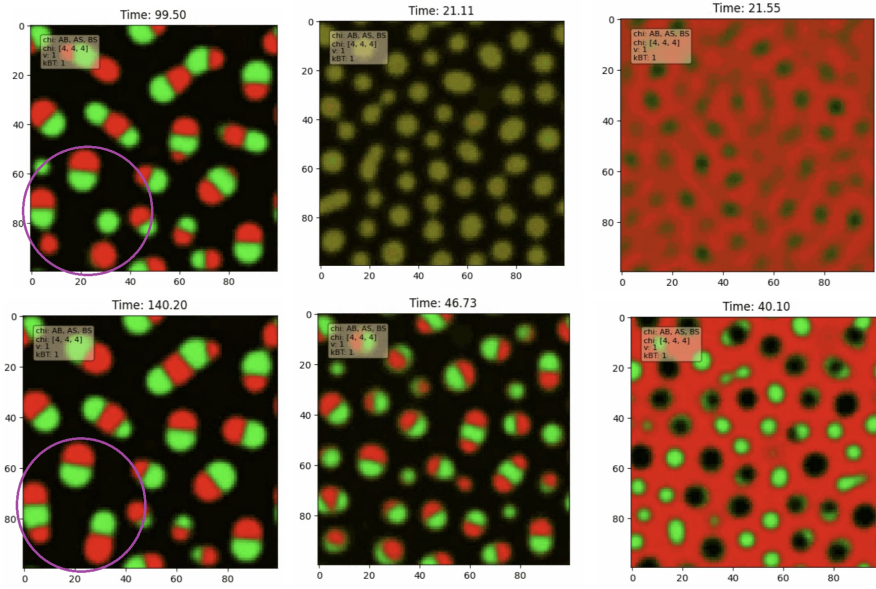


**Figure 2:** Snapshots of animations of species A (red) and B (green) phase separating on a background of solvent S (black), showing Ostwald-ripening (purple circle) and coalescence (blue circle). Graphs show mean radius over time with fits over shorter timescale (left) and longer timescale (right).

### 3.2 Testing surface tension and symmetry

A more qualitative test of the spatial dynamics is to observe the droplet behavior when all 3 species (A, B and solvent) phase separate from each other. We expect droplets of different species to stick to each other as this minimizes surface area, thus minimizing the surface tension energy.

In the case of two particle species a certain mean concentration is required to create new droplets in a well-mixed environment. The regime where new droplets emerge is called the nucleation regime (Weber et al. et al., 2019) [7]. Thus we would expect that if concentrations of species A and species B are very low they shouldn't phase separate from the solvent individually, however if their combined concentrations are large enough to nucleate new droplets we should see that initially two phases emerge; one consisting of solvent and one which is A and B mixed. Because the concentrations of A and B inside this second phase are large, they will eventually phase separate,

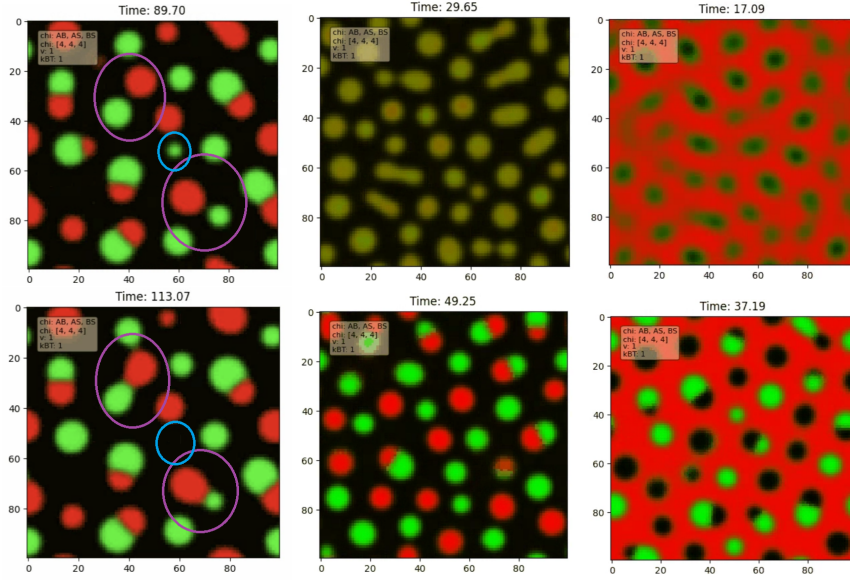


**Figure 3:** The images show the dynamics when all 3 species, A (Red), B (green), and solvent (Black) separate from each other. The top row shows an earlier timestamp whereas the lower row shows the corresponding systems at a later time. The left most column shows droplets sticking together due to surface tension. The other two columns show that first two major phases emerge. the smallest one of these split into two smaller phases at a later time. The center column is with mean volume fractions:  $[\bar{\phi}_A; \bar{\phi}_B; \bar{\phi}_S] = [0.17; 0.17; 0.66]$ . In the rightmost column the mean volume fractions are  $[\bar{\phi}_A; \bar{\phi}_B; \bar{\phi}_S] = [0.66; 0.17; 0.17]$ . Which is why they look similar just with red and black switched. Note that the lower image of the center column mainly contains droplets that are divided through the middle into two smaller droplets of different species, whereas the rightmost image almost only contains pure droplets of either species.

resulting in 3 phases. If all interaction parameters are equal ( $\chi_{AB} = \chi_{BC} = \chi_{AC}$ ) we would also expect symmetry in the sense, that if the concentrations of solvent and B are low, the initial 2 phases should be a phase of A and a phase in which B and the solvent are mixed. Fig. 3 shows how the simulation reflects this behavior. However, we notice that the symmetry is slightly off, as the simulation with high mean concentration of species A results in less droplets that "are split through the middle" as opposed to pure droplets consisting of only one particle species, this asymmetry is problematic, as the 3 particle species should be interchangeable in the simulation. However, the tendency to first split into two phases and then later turn into three seems to be happening in both cases.

### 3.3 Numerical error due to asymmetric representation of chemical potential of exchange

We notice a slight asymmetry when simulating at high A-concentration vs. high solvent-concentration in fig. 3. We realized that this was due to the definition of the



**Figure 4:** Shows the equivalent simulations of figure (3) with improved spatial dynamics. The snapshots are not taken at the same timestamps for the various simulations, as the random initial grids influence when the early nucleation greatly, however the  $\Delta t$  between snapshots is similar. Note in the leftmost column the droplet dissipating in the blue circle and the droplets in the purple circles sticking together. Notice that the center and rightmost column that the plots that both simulations start by separating into two phases and later separate into 3 phases. Note that the ratio between pure droplets, and droplets divided through the middle are similar in the lower row of the center and rightmost column.

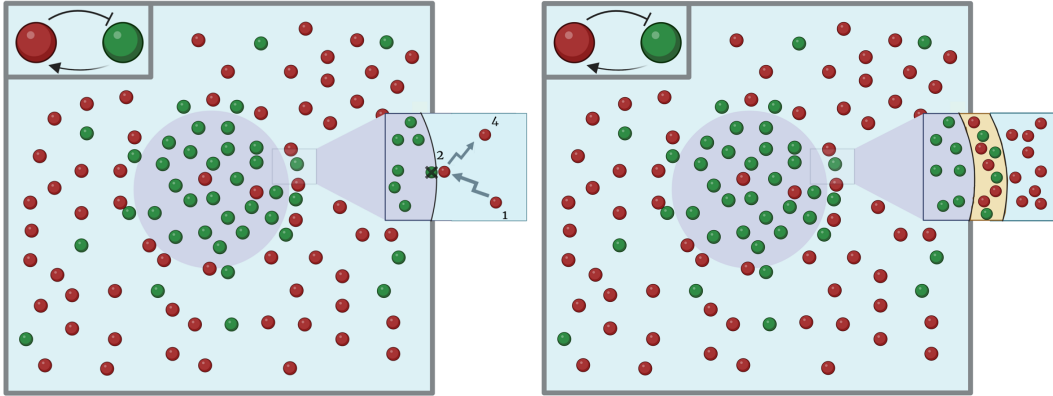


chemical potentials. The chemical potentials of exchange, eqs. (2) & (3), work well for analytic purposes, as A and B interact indirectly through the solvent. However, when we simulate it discretely, the solvent is updated at each time step according to the potential  $\mu_S = -\mu_1 - \mu_2$ , which means that it gets updated on average twice as much as A and B concentrations. Another problem is that the indirectness between A and B means that A only experiences change in B with one time step of delay. To fix this an extra chemical potential was added. the chemical potential of exchange between species A and species B.  $\mu_{AB} = \mu_A - \mu_B$ . In this notation the former chemical potentials of exchange would be denoted  $\mu_A \hat{=} \mu_{AC}$  and  $\mu_B \hat{=} \mu_{BC}$ .  $\mu_{AB}$  is added to the former chemical potentials such that the chemical potentials used in the simulation are:

$$\begin{aligned}\mu_{A-complete} &= \mu_{AC} + \mu_{AB} \\ \mu_{B-complete} &= \mu_{BC} - \mu_{AB}\end{aligned}\tag{15}$$

In fig. 4 it is shown that the system now behaves symmetric when we swap the concentration of A or B for concentration of solvent. It is seen by the center and rightmost columns being similar with the red color taking the place of black in the rightmost column. It is also clear that this doesn't mess with the surface tension and Ostwald ripening as droplets still stick to each other and small droplets dissipate. We now have a working model for spatial simulations, and move on to an analytical approach, to investigate how we expect this simulation to behave.

### 3.4 Feedback with single droplet - 3 models



**Figure 5:** The left image is an illustration of first passage reaction. 1 is the particle diffusing. 2 is when it finds its target. 3 is after the reaction is over and the particle diffuses away again. The right image is an illustration of surface reaction. Here the yellow area is where they are assumed to be well mixed.

We now take a step back and consider the more general case with  $N$  different species. With  $c_N \rightarrow c_1$  and  $c_i \rightarrow c_{i+1}$  otherwise. We propose 3 different ways to

model this: An unaffected model, a first passage model and a surface reaction model. Schematics of the latter two are displayed in fig. 5.

Without phase separation with the negative link being a Smoluchowski rate reaction [1], the dynamics are as follows:

$$\begin{aligned}\frac{dc_1}{dt} &= a_1 - b_1 c_1 c_N \\ \frac{dc_i}{dt} &= a_i c_{i-1} - b_i c_i, \quad i \neq 1\end{aligned}\tag{16}$$

Defining  $\prod a_i = A$  and assuming  $b_i = 1 \forall i$  gives fixed points:  $c_i^* = \frac{\prod_{j=1}^i a_j}{\sqrt{A}}$ . The characteristic polynomial of the Jacobian in the fixed point becomes:

$$(\lambda + \sqrt{A})(\lambda + 1)^{N-1} + \sqrt{A} = 0\tag{17}$$

Letting  $A = 1$  gives solutions of the form  $\lambda = -1 + (-1)^{\frac{1}{N}}$ . This rules out the possibility of sustained oscillations around the fixed point, as the real part of the eigenvalue never changes sign. Thus a Hopf bifurcation isn't possible.

If we then on the other hand restrict ourselves to the case of 2 species, and assume  $c_i \in [0; 1]$  which would be the case when looking at volume fractions, we get  $\lambda = \frac{-1 - \sqrt{A} \pm \sqrt{1 + A - 10\sqrt{A}}}{2}$ . However,  $c_2 < 1 \Rightarrow \sqrt{A} < 1$ , so the real part of the eigenvalues are always negative. For  $A \rightarrow 1$  the imaginary part increases but the real part moves further away from zero, so perturbations will decay faster with a higher frequency. When  $A \rightarrow 0$ , the imaginary parts disappear and we are left with two negative, real eigenvalues ( $\lim_{A \rightarrow 0} \lambda_+ = 0$ , but  $\lambda_+ < 0$  for  $A \in [0, 1]$ ).

### 3.4.1 First passage rate

The previous section didn't take into account phase separation. If phase separation takes place without chemical reactions, a system will after some time reach equilibrium, consisting of two phases, one phase takes the shape of a droplet surrounded by the other phase. We will now investigate the dynamics in the case of one big droplet, when chemical reactions are occurring. In the case of strong phase separation between two chemicals, they will only be allowed to react with each other on the surface of this droplet. If reactions on the surface occur immediately upon contact, the reaction rate occurs at the first passage rate of a particle reaching the surface of the droplet. In this case a droplet of species 1 can be considered absorbing sphere of radius  $r_1$ . We choose a coordinate system such that the droplet is centered at the origin.

In that case the boundary conditions turn out as:  $c_2(R) = 0$  because species 2 is "absorbed upon impact" and  $\lim_{r \rightarrow \infty} c_2 = \bar{c}_2$  because the concentration of species should approach the mean concentration when moving away from the droplet. The

Fokker-Planck diffusion equation for a freely diffusing particle in a radially symmetric concentration field reads:

$$\frac{dc_2}{dt} = D \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial c_2}{\partial r} \right) \quad (18)$$

The steady state solution to this, is:

$$c_2(r) = \begin{cases} -\frac{\bar{c}_2 r_1}{r} + \bar{c}_2 & \text{for } r \geq r_1 \\ 0 & \text{for } r < r_1 \end{cases} \quad (19)$$

For a freely diffusing particle species (species 2) the flux through a surface is [10]:

$$J_2 = - \int_{surface} D \frac{dc_2}{d\vec{n}} dA \quad (20)$$

Where D is the diffusion constant. Evaluating the integral on any sphere bigger than the droplet gives:

$$J_2 = 4\pi D r_1 c_2 \quad (21)$$

Which then gives us the amount of particles moving towards the surface, i.e. the reaction rate.

If the phase separation is strong, the radius of the droplet grows as  $r_1 = \nu^{\frac{1}{3}} N_1^{\frac{1}{3}} = \left( \bar{\phi}_1 V \right)^{\frac{1}{3}} = \left( \nu \bar{c}_1 V \right)^{\frac{1}{3}}$ , so the resulting reaction rate becomes:  $J_2 = 4\pi D (\nu V)^{\frac{1}{3}} c_2 c_1^{\frac{1}{3}}$ .

Letting species 1 phase separate from species N, setting  $b_1 = 4\pi D (\nu V)^{\frac{1}{3}}$  and incorporating this into eq. (16)

$$\begin{aligned} \frac{dc_1}{dt} &= a_1 - b_1 c_1^{\frac{1}{3}} c_N \\ \frac{dc_i}{dt} &= a_i c_{i-1} - b_i c_i, \quad i \neq 1 \end{aligned} \quad (22)$$

Letting  $b_i = 1 \forall i$  leads to fixed point to being:

$$c_i^* = \frac{\sqrt{a_1}}{(A)^{\frac{3}{4}}} \prod_{j=1}^i a_j \quad (23)$$

And the characteristic polynomial of the Jacobian at the fixed point turns out as:

$$\left( \lambda + \frac{1}{3} A^{\frac{3}{4}} a_1^{-\frac{1}{2}} \right) (\lambda + 1)^{N-1} + A^{\frac{3}{4}} a_1^{-\frac{1}{2}} = 0 \quad (24)$$

If analogously to the well-mixed case, we assume  $A = 1$  and restrict ourselves to the case of  $a_1 = \frac{1}{9} = 0.\bar{1}$  the eigenvalues turn out to be:

$$\lambda = \left( -\frac{1}{9} \right)^{-\frac{1}{2N}} - 1 \quad (25)$$

This opens for the possibility of sustained oscillations around the fixed point for  $N > 5$ , which wasn't the case for the well-mixed case no matter what  $a_1$  was chosen. Turning our attention to the case of  $N = 2$ , this results in eigenvalues:

$$\lambda = \frac{-1 - \frac{1}{3}A^{\frac{3}{4}}a_1^{-\frac{1}{2}} \pm \sqrt{1 + \frac{1}{9}A^{\frac{3}{2}}a_1^{-1} - \frac{16}{3}A^{\frac{3}{4}}a_1^{-\frac{1}{2}}}}{2} \quad (26)$$

The real part is always negative as for  $A^{\frac{3}{4}}a_1^{-\frac{1}{2}} \rightarrow \infty$  the square-root approaches  $\infty$  at the same rate as the  $\frac{1}{3}A^{\frac{3}{4}}a_1^{-\frac{1}{2}}$  in front, so  $\lim_{A^{\frac{3}{4}}a_1^{-\frac{1}{2}} \rightarrow \infty} \lambda_+ = -\frac{1}{2}$ . As  $A^{\frac{3}{4}}a_1^{-\frac{1}{2}} \rightarrow 0$  the eigenvalues show the same tendency as in the well-mixed case for  $A \rightarrow 0$

So even though phase separation should open the possibility of sustained oscillations, it is not the case, when only including 2 species, according to the first passage rate model.

### 3.4.2 Reactions on droplet surface

The phase separation of species 1 and species  $N$  means that there will be a region on the surface of the droplet, where species 1 and  $N$  co-exist, and where the degradation of species 1 by species  $N$  takes place. In the limit of slow chemical reactions, the thickness of this border region is governed by spatial interactions and depend mainly on the total concentration of the phase separating species  $\bar{c}_1 + \bar{c}_N$  and the Flory-Huggins interaction parameter  $\chi_{1,N}$ , but is largely invariant to droplet size. Fig. 6 shows that as the radial interaction profile on the surface of a small droplet is almost identical to the one for a large droplet at steady state. The volume of the shell around a droplet, in which the degradation process is allowed due to both species being present, scales with the surface area of the droplet. We therefore propose a model where the degradation of  $c_1$  is proportional to the droplet surface area. The surface area is in the case of strong phase separation given by:  $4\pi r^2 = 4\pi\nu^{\frac{2}{3}}c^{\frac{2}{3}}$  giving a reaction volume:  $l \cdot 4\pi\nu^{\frac{2}{3}}c^{\frac{2}{3}}$  where  $l$  is a characteristic thickness of the interface. Using this in eq. (16) gives:

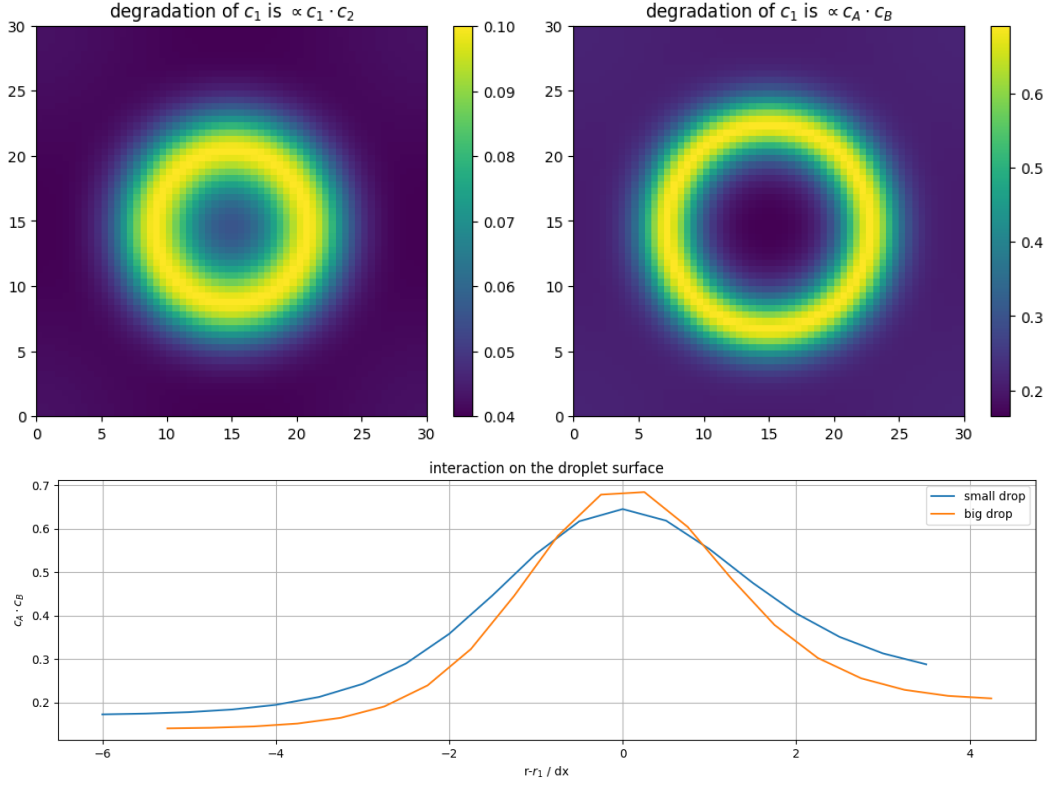
$$\begin{aligned} \frac{dc_1}{dt} &= a_1 - b_1 c_1^{\frac{2}{3}} c_N \\ \frac{dc_i}{dt} &= a_i c_{i-1} - b_i c_i, \quad i \neq 1 \end{aligned} \quad (27)$$

With fixed points at  $b_i = 1$ :

$$c_i^* = \left(\frac{a_1}{A^3}\right)^{\frac{1}{5}} \prod_{j=1}^i a_j \quad (28)$$

The characteristic polynomial of the Jacobian at the fixed point becomes:

$$\left(\lambda + \frac{2}{3}a_1^{-\frac{1}{5}}A^{\frac{3}{5}}\right)(\lambda + 1)^{N-1} + a_1^{-\frac{1}{5}}A^{\frac{3}{5}} = 0 \quad (29)$$



**Figure 6:** The upper two images show a heatmap of the interaction  $c_A \cdot c_B$  normalized according to total concentration  $c_A + c_B$ . The heatmaps show slices through the center of two 3-dimensional droplets of different sizes. The bottom plot shows  $c_A \cdot c_B$  plotted versus the radial coordinate in units of  $dx$ , the offset on the  $r$ -axis is such that the peaks line up. This is also normalized according to total concentration.

If, analogously to the other models we assume that  $A = 1$  and set  $a_1 = \frac{2^5}{3^5} \approx 0.132$  we get solutions of the form:

$$\lambda = \left(-\frac{2}{3}\right)^{-\frac{1}{N}} - 1 \quad (30)$$

This can result in the real part of the eigenvalue being positive thus making sustained oscillations possible for  $N \geq 13$ .

We look at the case of  $N = 2$  which yields the following eigenvalues:

$$\lambda = \frac{-1 - \frac{2}{3}a_1^{-\frac{1}{5}}A^{\frac{3}{5}} \pm \sqrt{1 + \frac{4}{9}a_1^{-\frac{2}{5}}A^{\frac{6}{5}} - \frac{16}{3}a_1^{-\frac{1}{5}}A^{\frac{3}{5}}}}{2} \quad (31)$$

This shows similar dynamics as the first passage model. Here the quantity in question becomes  $a_1^{-\frac{1}{5}}A^{\frac{3}{5}}$ , which like  $a_1^{-\frac{1}{2}}A^{\frac{3}{4}}$  grows in  $A$  and decays in  $a_1$ .

### 3.4.3 Numerical analysis of the two proposed models

None of the three models result in oscillations when  $N = 2$  however we expect a slight tendency to oscillate for a longer time, as sustained oscillations are in general expected, when  $N$  increases. A change in the stable fixed points is also expected, as all three models predict different fixed points. Simulating the three systems with randomized  $a$ -values, and  $A = 1$  results in behavior as observed in fig. 7.

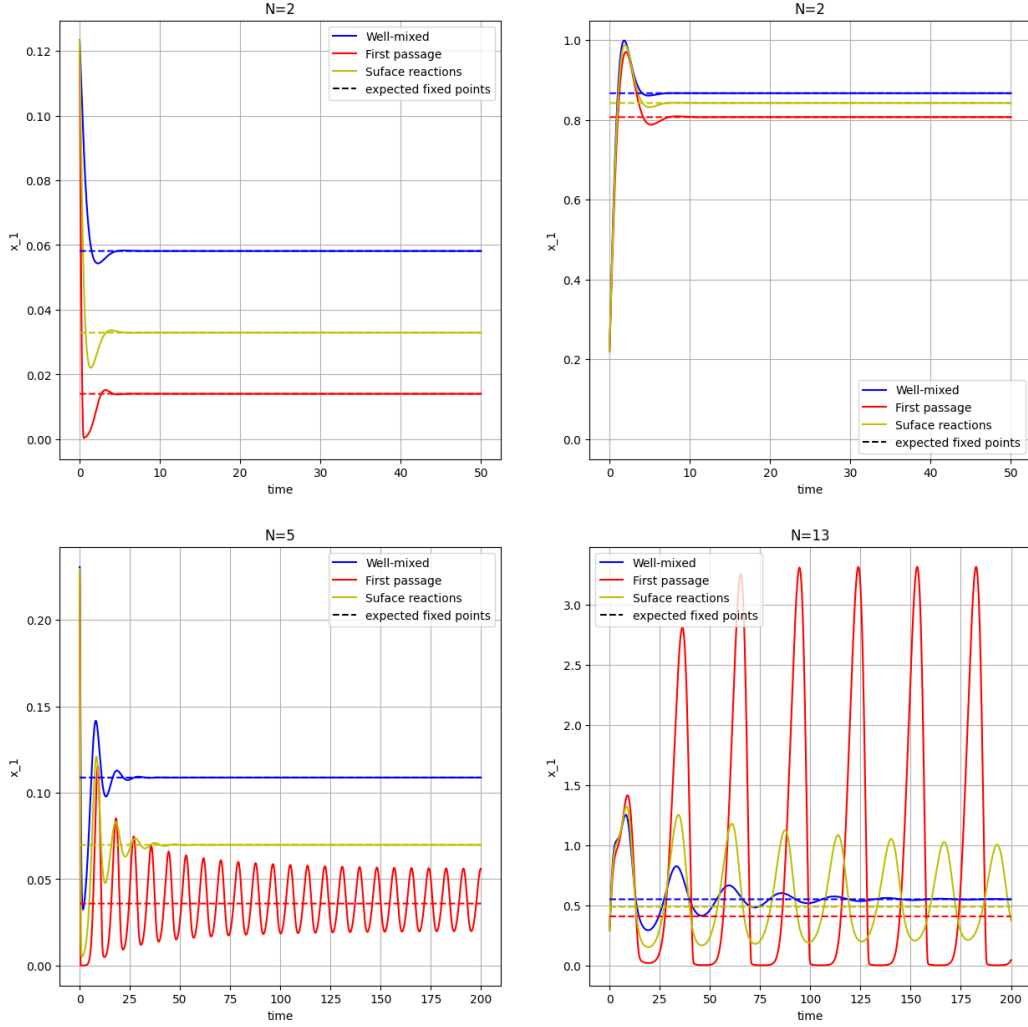
The first passage model and the surface reactions model display slightly longer oscillations than the well-mixed case for  $N = 2$ , although hardly any are observed. We also see that the fixed points change according to our predictions. In the case of  $N = 5$  and  $N = 13$ , the first passage model and the surface reactions model respectively, start to create sustained oscillations.

## 4 Analysis

### 4.1 Comparing spatial simulations to analytical models

The three analytical models exhibit very little oscillations when including only two particle species; for this reason we have chosen to approach the issue of checking their respective validity by looking at where the fixed point occurs in phase space, as all three predict three different fixed points. Even though the effective  $a_2$ - and  $b_i$ -values depend on system size,  $\chi$ -values, and diffusion rates, we should be able to determine which model best fits the system by only varying  $a_1$ , as the three models display different proportionality between the fixed points  $x_1^*$  and  $a_1$ , as seen in table 2.

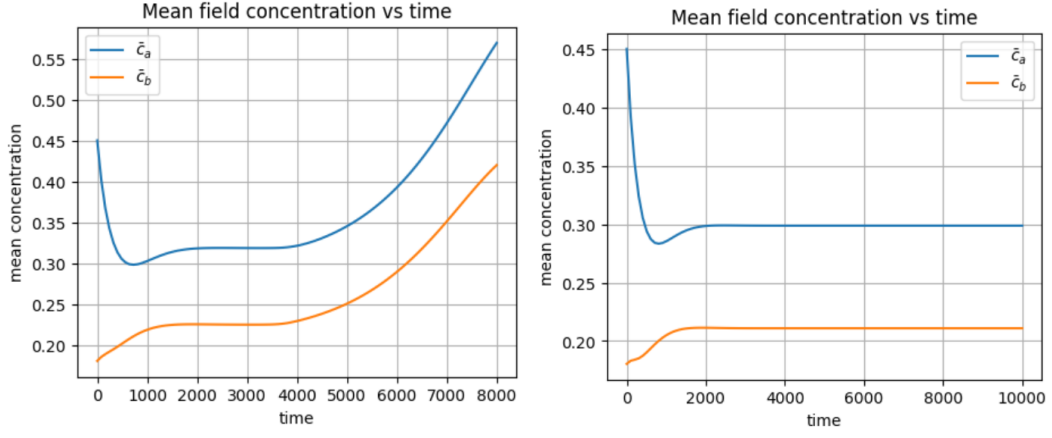
Running the spatial simulation resulted in mean concentrations like the one shown in fig. 9. We clearly see concentrations stabilizing after some time. By running the



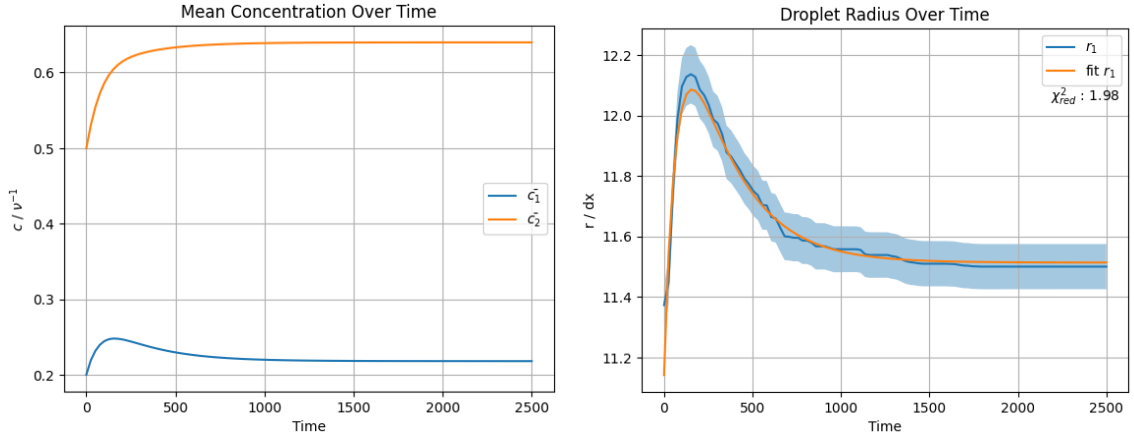
**Figure 7:** The four graphs show simulations of  $x_1$ -concentration using the three models: Well-mixed/mass-action (Blue), first passage (Red), Surface reactions (yellow). The expected fixed points are indicated with dashed lines. The  $a$ -values are randomized which is why the two graphs where the number of nodes  $N = 2$  look different from each other. Note that the first passage model displays sustained oscillations for  $N \geq 5$  and the surface reactions does so for  $N = 13$

| model              | $x_1^*$                     |
|--------------------|-----------------------------|
| No spatial impact  | $\propto a_1^{\frac{1}{2}}$ |
| first passage rate | $\propto a_1^{\frac{3}{2}}$ |
| surface reactions  | $\propto a_1^{\frac{6}{5}}$ |

**Table 2:** Expected dependency of  $x_1^*$  on  $a_1$



**Figure 8:** Simulations run with the chemical reactions from (8). Differing only slightly in  $a_1$ . The simulation shown on the right results in well mixed and the left diverges until incompressibility is no longer maintained and the simulation breaks

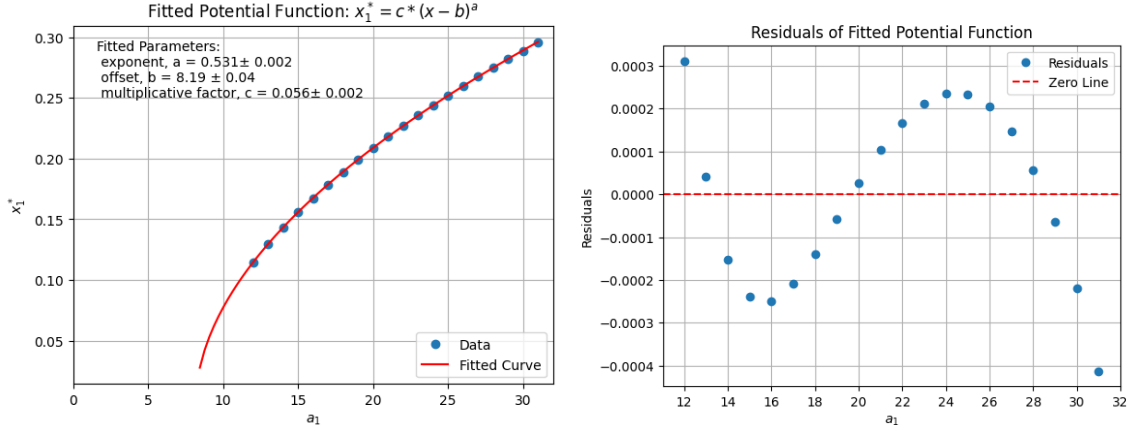


**Figure 9:** Running the spatial simulation using (10) for chemical reactions with parameters  $[a_1, b_1, a_2, b_2] = [0.35; 1; 2; 0.1]$ . The left-hand panel shows the mean-field concentrations. The right-hand panel shows the droplet radius. The error on the droplet radius is derived by assuming half of the pixels on the surface of the droplet might be mischaracterized as either part of or not part of the droplet. The fit is of the assumption:  $r_1 \propto \bar{c}_1^{-\frac{1}{3}}$  which seems to agree quite well with data.



simulation, while varying  $a_1$ , but keeping all other parameters constant we were able to fit how the fixed point concentration depends on  $a_1$ .

Fig. 10 shows how a potential function fits the fixed-point concentration of  $x_1$  versus  $a_1$ . It is clear that the fixed point scales with a power  $a < 1$  with a quite remarkable offset along the  $a_1$  axis. Furthermore the residuals show a systemic offset following what looks like a third order term, suggesting that a potential function in any case isn't right for describing the system.



**Figure 10:** Left: Simulated fixed-points of  $x_1$  concentration, fitting a potential function quite well with exponent  $a = 0.531 \pm 0.002$ . The results stem from simulations where errors were estimated by varying the tolerance when choosing time-steps. This however resulted in tiny errors. Right: Residuals of the fit. There seems to be a systemic error suggesting an incomplete theory.

## 5 Discussion & Outlook

### 5.1 Spatial dynamics

Figure 2 shows clear signs of coalescence and Ostwald-ripening. To get better idea of whether or not the coarsening of droplets is behaving as we would expect, we look at the mean droplet radius over time. The Lifshitz-Slyozov scaling law (7) tells us the mean radius should be proportional to  $\sqrt[3]{t}$ . This seems to only be the case for the short time scale dataset. This could be because the derivation of the scaling law assumes a very large number of droplets in a very large system. On the shorter time scale the droplet size and count does not exceed the limitations of this assumption. At longer timescales these limitations are exceeded as the system reaches a point where the number of droplets are low enough to halt the droplet growth rate. For this reason we tested the exponential fit and by looking at the  $\chi_{red}^2$  values saw that this is better at describing the behavior of the system over timescales where the number of droplets tends towards 1. Overall the spatial dynamics seems to be working as intended.

## 5.2 Chemical reactions

The full spatial simulation revealed multiple effects not accounted for in the analytical models. A discussion of the difference between the two approaches is thus essential. First of all, the chemical reactions we simulated, that are described in eq. (10), were different to the chemical reactions used in the analytical derivations which are the ones described in eq. (8). However, the latter exhibits either unstable diverging growth of concentrations, which means the spatial dynamics break down or the droplet vanishes and the result is a well-mixed environment, see figure 8. This behavior in itself warrants an investigation. We chose to focus our energy on investigating the behavior of simulating eq. (10) as this gave stable simulations that actually were comparable to predictions. In the limit  $c_s \gg c_2 + c_1$  eq. (10) exhibited similar instability as eq. (8) but when diverging quickly reached a regime where  $c_s \sim c_2 + c_1$ .

Inside the cell, proteins are produced by ribosomes translating mRNA, which is transcribed from the DNA. The activation of a certain gene on the DNA, and thus production of the corresponding protein, might be repressed or promoted by different transcription factors. Thus, protein production does not occur at the same location as the promoting agent is activating a gene. The assumption of local mass-action in the spatial simulation, does not take this into account as production in the simulation happens exactly in the same spot, where the promoter is present. Especially in the case of eq. (10) this becomes an issue due to its non-linear nature, as production of the given protein doesn't scale with the overall concentration as phase separation means that the promoting agent would accumulate making the solvent concentration in that region smaller. When simulating using eq. (8) locally this isn't a problem as constant production rate in each point means constant production across the entire grid, and linear dependency of only one species also means that the total production scales linearly with the mean concentration.

There is however another issue with the local mass-action assumption. Given a feedback loop where B promotes A and A inhibits B, and if all reactions happen in the same spot, diffusion times vanish which clashes with the basic assumption of the first passage model.

Another reason the first-passage model and the simulation differ is the assumption that a particle disappears upon impact with the droplet. In the model the droplet is degraded by chemicals in the surrounding environment, so impact with the droplet doesn't in any way degrade the surrounding chemicals. In a biological system this makes sense even though the free particle degrades the particles of the droplet. There might be a transient state where the degrading agent binds to the target, however the simulation does not encompass this. This means that instead of a particle only being responsible for one reaction upon impact, it is allowed to react until it either diffuses out again or vanishes due to decay. Arguably the particle would diffuse out pretty quickly as the phase separation would push it outwards which then introduces another discrepancy with the first passage model. The model assumes free diffusion.

This is per definition not the case, however this isn't as big of an issue as the other points. As a particle diffuses freely inside each phase and its motion only gets limited when getting close to the phase interface. At that point it has already almost reached its target, so this shouldn't present a huge problem.

### 5.3 Droplet size scaling law

The two proposed models assume that the volume of a droplet of species A scales proportionally to  $\bar{c}_A$ , however in the case of 2 particle species that separate from each other the scaling is more similar to  $\frac{c_A}{c_A+c_B}$ . For  $c_A \ll c_B$  this isn't a problem as  $c_B$  in that case dominates the denominator. In a physical system like p53 this would be the case as p53 makes up a tiny part of the entire cell, so the ratio of p53 to chemicals from which it separates would be quite small. In the simulation this isn't the case as species A ends up being responsible for a significant portion of the available number of particles.

In fig. 9 it is seen that the assumption in that particular case holds pretty well. By simulating species A phase separating from both B and the solvent this issue should however be circumvented completely as the droplet volume in this case should scale proportionally to  $\frac{c_A}{\bar{c}_A+\bar{c}_B+\bar{c}_S} = \bar{c}_A\nu = \bar{\phi}_A$ . Whereas solvent is not present at all inside the droplet. Species A would thus only be produced outside the droplet given the reactions described in eq. (10) and a great part of this A would then be degraded before even reaching the droplet, so the assumption that the degradation mainly occurs on the droplet surface would not hold.

### 5.4 Further exploration

None of the models exhibit sustained oscillations in the case of two particle species, no matter which of the analytical models we chose. Thus, more species are required to achieve this. A clear path would be to generalize our model for  $N > 2$ . This would not be all that hard to implement, as the Flory-Huggins potential generalizes quite simply. It would, however complicate matters in terms of concentrations needing to be very small, which first of all would make the log-term of the Flory-Huggins potential very volatile meaning very short time steps would be required, and also a certain minimum concentration is to be met for the Flory-Huggins potential to result in phase separation[7]. The latter might be solved by choosing the right interaction parameters,  $\chi_{i,j}$  such that groups of particles phase separate, or by simply choosing large enough  $\chi_{i,j}$ .

## 6 Conclusion

To summarize; we investigated how negative feedback loops in chemical kinetics are affected by LLPS. To do this we used both analytical models and numerical simulations. The simulations were done by constructing a program that allowed us to solve the differential equations following the Flory-Huggins potential by applying finite difference method and incorporating chemical kinetics. Analytically we tested three different models, one assuming a well mixed system, and two assuming phase separation. One model with phase separation assumed instant reactions upon impact leading to a first passage reaction rate. The other assumed a surface layer between phases where reactions continually occur. The well-mixed case does not display sustained oscillations, whereas the two phase separated models displayed sustained oscillations when using respectively  $N \geq 5$  and more than  $N \geq 13$  particle species in the feedback loop. Due to the large number of particles required to validate this behavior and our program being limited to two particle species, we chose to approach the issue of validation by looking at how the fixed point concentrations depended on reaction rates, as all three models predicted different dependencies, which are lined up in table 2.

Our simulations displayed expected physical phenomenons, such as Ostwald Ripening and Coalescence conforming nicely to classic theory and validating the spatial dynamics. Discrepancies were found when simulating with chemical reactions and comparing it to the analytical models. The behavior of the system when simulating the chemical kinetics from eq. (8) was unexpectedly unstable. The simulations using eq. (10) produced results more in line with the expectations, however the resulting fixed point concentrations did not match any of the analytical predictions. We discuss a variety of possible reasons for these discrepancies, mainly the assumption of local mass-action, which in the context of a biological system isn't always the case as specific processes occur certain places within the cell.

Although we were unable to arrive at a model able to accurately describe the impact phase separation has on negative feedback loops of chemical kinetics, we believe further exploration with a greater number of species might possibly yield better results, as this would allow for the possibility of sustained oscillations, and it would to some extent also be able to reflect the non-local nature of a biological system.

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## 8 Appendix

### 8.1 A - Flory-Huggins derivation

Suppose all the molecules are organized in a grid. If we have 3 molecule species A, B and C, with  $M = N_A + N_B + N_C$  being the total number of molecules. All molecules have the same volume  $\nu$ .

The energy associated with neighbor interactions can be denoted  $e_{ij}$  where  $i, j \in \{A, B, C\}$ . Eg. if molecule A and molecule B are next to each other, the corresponding energy will be  $e_{AB}$ . The neighbor interaction energy matrix is symmetrical such that  $e_{ij} = e_{ji}$ .

If we now have a grid of  $N$  grid points. We now represent the occupation of a given grid-point with the vector  $\sigma$ . The set  $\{\sigma\}$  denotes the collection of the  $\sigma$ -values of all grid points. This vector has the size of the number of particles in the system (in this case 3) so each particle will be represented by the regular Euclidean unit vectors. The energy of a given arrangement of molecules will then be given by:

$$H(\{\sigma\}) = \frac{1}{2} \sum_{(m,n)} \sigma_m^T \times \mathbf{e} \times \sigma_n \quad (32)$$

Summing over all neighbor pairs  $(m, n)$ , the factor  $\frac{1}{2}$  takes care of double counting.

The statistical weight of a given arrangement is then given by:  $Z(\{\sigma\}) = e^{\frac{H(\{\sigma\})}{k_B T}}$  so the statistical weight of  $N_A$  A molecules and  $N_B$  B molecules becomes:

$$Z(N_A, N_B, N_C) = \sum_{\Omega} e^{\frac{H(\{\sigma\})}{k_B T}} \quad (33)$$

Here the sum over  $\Omega$  denotes all the possible ways  $\{\sigma\}$  can be arranged. This sum becomes increasingly hard to calculate with increasing  $M$ , so to approximate this we employ the mean-field approximation. The mean field approximation works by letting each grid point take the value of the mean field, so:  $\sigma = (\phi_A, \phi_B, \phi_C)$ . Here  $\phi_i = \frac{N_i}{M}$  is the volume fraction of species  $i$ . In addition, we will apply the continuum limit in which the finite grid turns into a continuum. In this limit we view the  $\phi_i(\vec{r})$ ,  $i = A, B, C$  as spatial functions, so we locally assume a well mixed state (mean-field approximation) but allow the volume fractions to change on a global scale.

From the partition function, the Helmholtz free energy,  $F$ , can be calculated, from which a range of physical properties are derived.

$$F = -k_B T \ln(Z) \quad (34)$$

We will focus on an incompressible system which makes sense for the grid assumption but also for simplicity, so the total volume is conserved:  $V = M\nu$ .

$$\mu_i = \left. \frac{\partial F}{\partial N_i} \right|_{T,V} = \left. \frac{\partial V \cdot f}{\partial M \cdot \phi_i} \right|_{T,V} = \left. \frac{\partial V \cdot f}{\partial \frac{V}{\nu} \cdot \phi_i} \right|_{T,V} = \nu \cdot \left. \frac{\partial f}{\partial \phi_i} \right|_T \quad (35)$$

Here  $f$  is the free energy density, and  $\mu$  is the chemical potential. To calculate this we need a more compact way of evaluating the free energy. Due to the mean field approximation eq. (34) reduces to:

$$F = -k_B T \ln |\Omega| + H(\{\sigma\}) \quad (36)$$

Now the magnitude of  $\Omega$  is just the number of ways we might arrange the molecules in the grid. This is done by applying the binomial coefficient, so:

$$|\Omega| = \binom{M}{N_A + N_B} \cdot \binom{N_A + N_B}{N_A} = \frac{M!}{(N_A + N_B)! \cdot N_C!} \cdot \frac{(N_A + N_B)!}{N_A! \cdot N_B!} = \frac{M!}{N_A! \cdot N_B! \cdot N_C!} \quad (37)$$

using Stirling's approximation this becomes:

$$\ln |\Omega| \approx N_A \ln\left(\frac{N_A}{M}\right) + N_B \ln\left(\frac{N_B}{M}\right) + N_C \ln\left(\frac{N_C}{M}\right) = \frac{V}{\nu} \cdot \left[ \sum_{i=A,B,C} \phi_i \ln \phi_i \right] \quad (38)$$

$$= \frac{1}{\nu} \int dr^3 \left[ \sum_{i=A,B,C} \phi_i \ln \phi_i \right] \quad (39)$$

In the last line we changed to the continuum limit where  $V = \int dr^3$ .

Now turning to the  $H$ -term of the free energy we have:

$$\begin{aligned} H(\{\sigma\}) &= \frac{1}{2} \cdot \sum_{(m,n)} (\sigma_{A1} \quad \sigma_{B1} \quad \sigma_{C1}) \begin{pmatrix} e_{AA} & e_{AB} & e_{AC} \\ e_{AB} & e_{BB} & e_{BC} \\ e_{AC} & e_{BC} & e_{CC} \end{pmatrix} \begin{pmatrix} \sigma_{A2} \\ \sigma_{B2} \\ \sigma_{C2} \end{pmatrix} \\ &= \frac{1}{2} \sum_{(m,n)} [e_{AA}\sigma_{Am}\sigma_{An} + e_{BB}\sigma_{Bm}\sigma_{Bn} + e_{CC}\sigma_{Cm}\sigma_{Cn} \\ &\quad + e_{AB}(\sigma_{Am}\sigma_{Bn} + \sigma_{Bm}\sigma_{An}) + e_{AC}(\sigma_{Am}\sigma_{Cn} + \sigma_{Cm}\sigma_{An}) + e_{BC}(\sigma_{Bm}\sigma_{Cn} + \sigma_{Cm}\sigma_{Bn})] \\ &= \frac{1}{2} \sum_{(m,n)} \left[ \sum_{i=A,B,C} e_{ii}\sigma_{im}\sigma_{in} + \frac{1}{2} \sum_{j \neq i} e_{ij}(\sigma_{im}\sigma_{jn} + \sigma_{jm}\sigma_{in}) \right] \end{aligned} \quad (40)$$

The sum over  $j \neq i$  takes care of the of-diagonal terms, the factor  $\frac{1}{2}$  takes care of double counting.

Now remembering that the system is incompressible and using  $2\sigma_{im}(1 - \sigma_{in}) = (\sigma_{im} - \sigma_{in})^2 - \sigma_{im}^2 - \sigma_{in}^2 + 2\sigma_{im}$  and  $(\sigma_{im} - \sigma_{in})(\sigma_{jm} - \sigma_{jn}) = \sigma_{im}\sigma_{jm} + \sigma_{in}\sigma_{jn} - \sigma_{im}\sigma_{jn} - \sigma_{jm}\sigma_{in}$  this becomes:

$$\begin{aligned} H(\{\sigma\}) &= \frac{1}{2} \sum_{(m,n)} \left[ \sum_{i=A,B,C} e_{ii} \left( \sigma_{im} - \frac{1}{2}(\sigma_{im} - \sigma_{in})^2 + \frac{1}{2}\sigma_{im}^2 + \frac{1}{2}\sigma_{in}^2 - \sigma_{im} \right) \right. \\ &\quad \left. + \frac{1}{2} \sum_{j \neq i} e_{ij} (\sigma_{im}\sigma_{jm} + \sigma_{in}\sigma_{jn} - (\sigma_{im} - \sigma_{in})(\sigma_{jm} - \sigma_{jn})) \right] \end{aligned} \quad (41)$$

Then applying the mean-field approximation we get (the sum becomes  $\frac{z}{\nu} \int dr^3$  with  $z$  being the average number of nearest neighbors of each particle e.g.  $z = 6$  for a regular cubic grid structure) and in the continuum limit  $\langle \sigma_i \rangle = \phi_i(\vec{r})$  and  $\langle \sigma_{im} - \sigma_{in} \rangle = \nu^{\frac{1}{3}} \nabla \phi_i(\vec{r})$ . The latter becomes clear if we consider that each neighbor interaction must happen in some direction and the mean rate of change in that direction must be the component of the gradient of the mean field in that direction, but as the gradient is a change in volume fraction per distance we need to multiply by the mean-distance between grid points, namely  $\nu^{\frac{1}{3}}$ . Combining all this gives:

$$H(\{\sigma\}) = \frac{1}{2} \frac{z}{\nu} \int dr^3 \left[ \sum_{i=A,B,C} e_{ii} \left( \phi_i - \frac{\nu^{\frac{2}{3}}}{2} \nabla \phi_i \cdot \nabla \phi_i + \phi_i^2 - \phi_i \right) + \frac{1}{2} \sum_{j \neq i} e_{ij} \left( 2 \cdot \phi_i \phi_j - \nu^{\frac{2}{3}} \nabla \phi_i \cdot \nabla \phi_j \right) \right] \quad (42)$$

Now using  $\phi_i - \phi_i^2 = \phi_i(1 - \phi_i) = \phi_i(\phi_j + \phi_k) = \phi_i \phi_j + \phi_i \phi_k$  and  $\nabla \phi_A + \nabla \phi_B + \nabla \phi_C = 0$

$$H(\{\sigma\}) = \frac{1}{2} \frac{z}{\nu} \int dr^3 \left[ \sum_{i=A,B,C} e_{ii} \left( \phi_i + \frac{\nu^{\frac{2}{3}}}{2} \nabla \phi_i \cdot (\nabla \phi_j + \nabla \phi_k) - \phi_i \phi_j - \phi_i \phi_k \right) + \frac{1}{2} \sum_{j \neq i} e_{ij} \left( 2 \cdot \phi_i \phi_j - \nu^{\frac{2}{3}} \nabla \phi_i \cdot \nabla \phi_j \right) \right] \quad (43)$$

Collecting terms proportional to  $\nabla \phi_i \cdot \nabla \phi_j^2$ ,  $\phi$  and  $\phi_i \phi_j$  yields:

$$H(\{\sigma\}) = \frac{1}{2} \frac{z}{\nu} \int dr^3 \left[ \sum_{i=A,B,C} e_{ii} \phi_i + \frac{1}{2} \sum_{j \neq i} \left( (2e_{ij} - e_{ii} - e_{jj}) \left( \phi_i \phi_j - \frac{\nu^{\frac{2}{3}}}{2} \nabla \phi_i \cdot \nabla \phi_j \right) \right) \right] \quad (44)$$

We define  $\chi_{ij} = \frac{z(2e_{ij} - e_{ii} - e_{jj})}{k_B T}$  (This is the Flory-Huggins parameter [7]). The different  $\chi$ 's quantize how much energy it takes to mix two chemicals. The greater the  $\chi$  the more the two chemicals will tend to demix.

Defining  $F_0 = M \cdot z \cdot (e_{AA} \bar{\phi}_A + e_{BB} \bar{\phi}_B + e_{CC} \bar{\phi}_C)$  where the bar denotes mean, is the energy of the system without taking mixing into account.

And the complete free energy can then be expressed as a function of only  $\phi_A$  and  $\phi_B$



when invoking incompressibility:

$$\begin{aligned}
F(\phi_A, \phi_B) = F_0 + \int dr^3 \frac{1}{\nu} & \left( k_B T \cdot \left[ \phi_A \ln \phi_A + \phi_B \ln \phi_B + (1 - \phi_A - \phi_B) \ln(1 - \phi_A - \phi_B) \right. \right. \\
& + \chi_{AB} \phi_A \phi_B + (\chi_{AC} \phi_A + \chi_{BC} \phi_B)(1 - \phi_A - \phi_B) \\
& \left. \left. - \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \nabla \phi_A \nabla \phi_B - \chi_{BC} |\nabla \phi_B|^2 - \chi_{AC} |\nabla \phi_A|^2 \right) \right] \right)
\end{aligned} \tag{45}$$

The system will try to minimize the free energy, i.e. minimize the integral, which is solved by the euler-lagrange equation. Thus the system will evolve such that chemical potentials given by the functional derivative of the functional  $f$  inside the integral will be minimized.

This results in the chemical potentials [5]:

$$\begin{aligned}
\mu_A = \nu \frac{\delta f}{\delta \phi_A} = \ln \frac{\phi_A}{1 - \phi_A - \phi_B} & + (\chi_{AB} - \chi_{BC}) \phi_B + \chi_{AC} (1 - 2\phi_A - \phi_B) \\
& + \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \Delta \phi_B - 2\chi_{AC} \Delta \phi_A \right)
\end{aligned} \tag{46}$$

$$\begin{aligned}
\mu_B = \nu \frac{\delta f}{\delta \phi_B} = \ln \frac{\phi_B}{1 - \phi_A - \phi_B} & + (\chi_{AB} - \chi_{AC}) \phi_A + \chi_{BC} (1 - \phi_A - 2\phi_B) \\
& + \frac{\nu^{\frac{2}{3}}}{2} \left( (\chi_{AB} - \chi_{BC} - \chi_{AC}) \Delta \phi_A - 2\chi_{BC} \Delta \phi_B \right)
\end{aligned} \tag{47}$$

## 8.2 B - Code in Github repository

